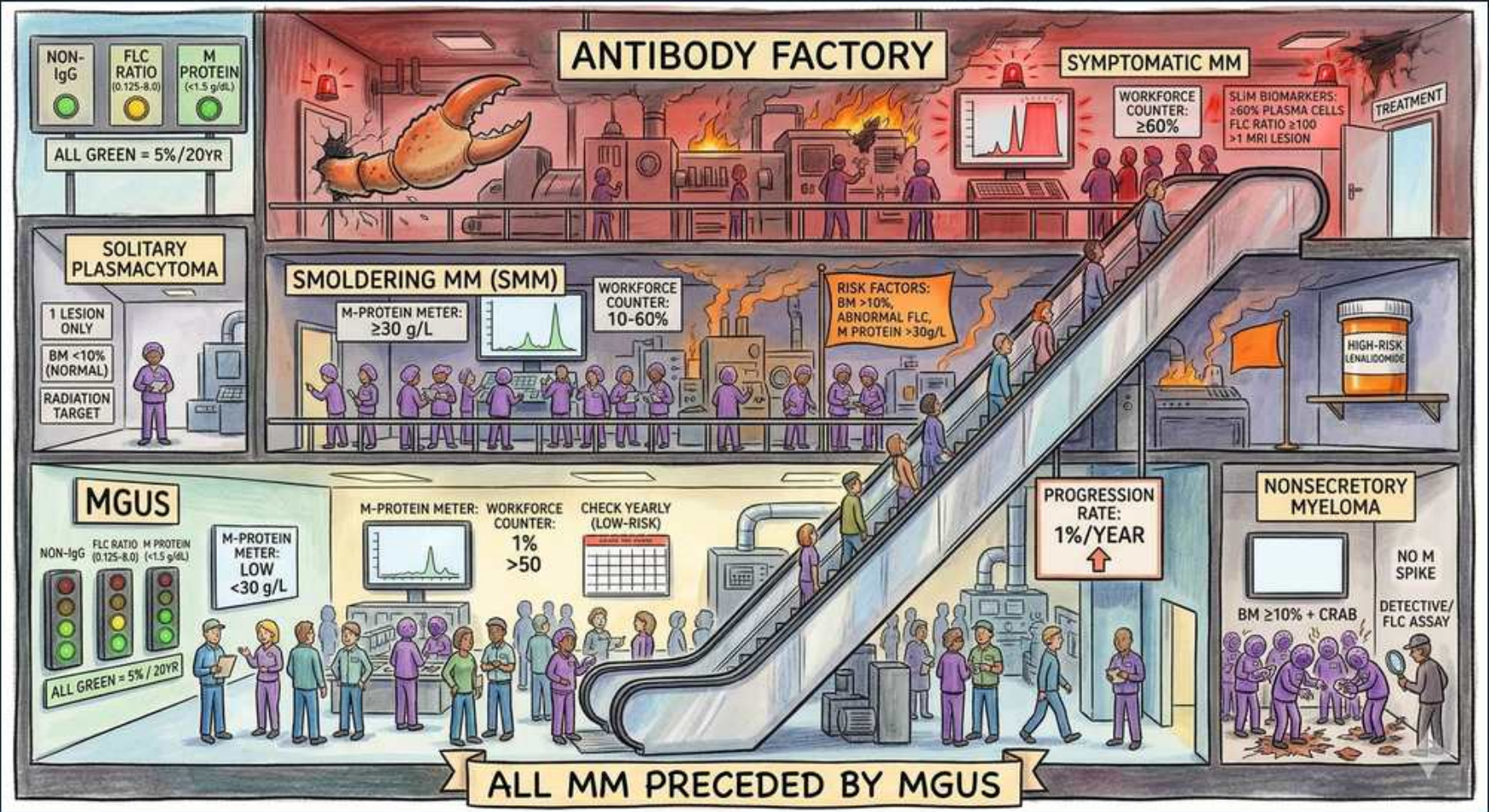


Plasma Cell — MGUS to Smoldering to Myeloma Spectrum



1. MGUS criteria

WHAT YOU SEE

MGUS lobby, M-protein meter LOW, less than 30 g/L

WHAT IT ENCODES

MGUS (monoclonal gammopathy of undetermined significance) requires ALL three: serum M-protein <3 g/dL (<30 g/L) AND clonal marrow plasma cells <10% AND NO CRAB/SLiM end-organ damage. It is a premalignant clone present in ~3% of people over 50 and ~5% over 70. Management is observation, not treatment — recheck SPEP/CBC/calcium/creatinine periodically.

BOARD TRAP

WRONG answer: starting myeloma therapy (e.g., lenalidomide/bortezomib) for MGUS. The discriminator is that MGUS has no end-organ damage by definition — ANY CRAB feature or SLiM biomarker reclassifies the patient as multiple myeloma, which is what gets treated; MGUS is only monitored.

2. MGUS plasma cells less than 10%

WHAT YOU SEE

WORKFORCE COUNTER 1% with greater than 50 sign and CHECK YEARLY

WHAT IT ENCODES

MGUS marrow clonal plasma cells must be <10% (often a marrow isn't even needed if M-protein is low and labs are clean). The clone progresses to myeloma or a related malignancy at a steady ~1% per year. Mnemonic: 'MGUS marches at 1% a year' — risk is lifelong, so surveillance never fully stops.

BOARD TRAP

WRONG answer: reassuring the patient that MGUS is benign and discharging them with no follow-up. The discriminator is the cumulative ~1%/year progression risk — MGUS needs ongoing surveillance (interval guided by risk stratification), not one-and-done reassurance.

3. MGUS all-green (no damage)

WHAT YOU SEE

ALL GREEN 5% / 20yr traffic lights at bottom of MGUS

WHAT IT ENCODES

MGUS lifetime progression is roughly 1%/year (~25% over a lifetime), and most patients never progress and die of other causes. Risk stratifies on three factors: non-IgG isotype, M-protein ≥1.5 g/dL, and abnormal serum free light chain ratio (0 vs 1–3 adverse factors). Low-risk IgG MGUS <1.5 g/dL with normal FLC ratio can have minimal follow-up.

BOARD TRAP

WRONG answer: treating all MGUS as equally low risk. The discriminator is the Mayo risk model — IgA/IgM isotype, M-protein ≥1.5 g/dL, and abnormal FLC ratio define high-risk MGUS warranting closer monitoring, sometimes with a baseline marrow and imaging.

4. Smoldering MM (SMM)

WHAT YOU SEE

SMOLDERING MM (SMM) banner, smoking factory

WHAT IT ENCODES

Smoldering multiple myeloma is the intermediate stage — greater tumor burden than MGUS but still NO CRAB and no SLiM biomarkers (so no end-organ damage). It is asymptomatic by definition and historically observed, though high-risk SMM is increasingly offered early lenalidomide on trials. Mnemonic: 'smoldering = fire is lit but the building isn't burning (no CRAB).'

BOARD TRAP

WRONG answer: treating all smoldering myeloma like active myeloma. The discriminator is the absence of CRAB/SLiM — standard care is close observation; only HIGH-RISK SMM (or development of a myeloma-defining event) prompts therapy, and early lenalidomide is largely trial/guideline-evolving.

5. SMM M-protein greater than or equal to 30 g/L

WHAT YOU SEE

M-PROTEIN METER greater than or equal to 30 g/L

WHAT IT ENCODES

SMM is defined by serum M-protein ≥3 g/dL (≥30 g/L) OR urine M-protein ≥500 mg/24h, OR marrow clonal plasma cells 10–60%, in the absence of CRAB and SLiM events. So it sits between MGUS (<3 g/dL and <10% cells) and myeloma (any defining event). Numbers separate the stages — burden up, but organs still intact.

BOARD TRAP

WRONG answer: assuming an M-protein ≥3 g/dL automatically means symptomatic myeloma needing chemo. The discriminator is end-organ status — without CRAB or a SLiM biomarker, a ≥3 g/dL spike is smoldering myeloma, which is observed, not immediately treated.

6. SMM plasma cells 10-60%

WHAT YOU SEE

WORKFORCE COUNTER 10-60%

WHAT IT ENCODES

SMM marrow clonal plasma cells fall in the 10–60% range. The upper bound matters: reaching ≥60% is a SLiM myeloma-defining event that diagnoses myeloma even without CRAB. Mnemonic for the 'Sixty' in SLiM: 60% plasma cells = the building is full, call it myeloma.

BOARD TRAP

WRONG answer: waiting for CRAB to appear before treating a patient whose marrow shows 65% clonal plasma cells. The discriminator is the SLiM '60%' rule — ≥60% marrow plasma cells defines myeloma now and is an indication to treat, regardless of intact end-organs.

7. SMM risk factors

WHAT YOU SEE

RISK FACTORS box: M-protein greater than 30g/L, abnormal FLC ratio

WHAT IT ENCODES

High-risk SMM is captured by the '20-2-20' model: M-protein >2 g/dL, marrow plasma cells >20%, and FLC ratio >20 — any 2 of 3 marks high risk for rapid progression (~50% at 2 years). These patients are candidates for early intervention/trials. Mnemonic: '20-2-20' = the three twenties of dangerous smoldering disease.

BOARD TRAP

WRONG answer: using only M-protein level to gauge SMM risk. The discriminator is the multi-factor 20-2-20 stratification (M-protein, marrow plasma cell %, and FLC ratio together) — high-risk SMM behaves almost like myeloma and warrants closer follow-up or clinical-trial therapy.

8. Symptomatic MM

WHAT YOU SEE

SYMPTOMATIC MM top floor, M-spike chart

WHAT IT ENCODES

Active multiple myeloma = clonal marrow plasma cells (usually $\geq 10\%$) PLUS at least one myeloma-defining event: any CRAB feature OR any SLiM biomarker. This is the stage that gets treated — typical induction is a triplet/quadruplet such as VRd (bortezomib + lenalidomide + dexamethasone) $\pm$ daratumumab, then autologous stem cell transplant in eligible patients. Mnemonic: 'a defining event turns the clone into a disease.'

BOARD TRAP

WRONG answer: requiring symptoms (bone pain, fatigue) before diagnosing myeloma. The discriminator is the 2014 IMWG update — SLiM biomarkers (60% plasma cells, FLC ratio ≥ 100 , >1 focal MRI lesion) define myeloma in still-asymptomatic patients so therapy can start before CRAB damage occurs.

9. MM workforce greater than or equal to 60%

WHAT YOU SEE

WORKFORCE COUNTER greater than or equal to 60%

WHAT IT ENCODES

Clonal marrow plasma cells $\geq 60\%$ is the 'S' of SLiM and independently defines multiple myeloma even with no CRAB, because progression to organ damage is nearly inevitable. It removes the old requirement to wait for kidney/bone/calcium injury. This is the 'Sixty percent' criterion.

BOARD TRAP

WRONG answer: classifying a patient with 62% marrow plasma cells but normal calcium/creatinine/Hgb as smoldering myeloma. The discriminator is the SLiM $\geq 60\%$ rule — that alone upgrades them to active myeloma requiring treatment.

10. SLiM biomarkers

WHAT YOU SEE

SLiM box: greater than or equal to 60% plasma cells, FLC ratio greater than or equal to 100, greater than 1 MRI lesion

WHAT IT ENCODES

SLiM myeloma-defining biomarkers: Sixty percent clonal marrow plasma cells, Light-chain (involved:uninvolved free) ratio ≥ 100 (with involved FLC ≥ 100 mg/L), and MRI with >1 focal lesion (each ≥ 5 mm). Any one defines multiple myeloma WITHOUT CRAB. Mnemonic: SLiM = Sixty / Light chains 100 / MRI lesions.

BOARD TRAP

WRONG answer: telling a patient with an FLC ratio of 120 and no CRAB that they only have smoldering disease. The discriminator is that any single SLiM biomarker (here, FLC ratio ≥ 100) is myeloma-defining and an indication to treat, even with normal calcium, kidneys, hemoglobin, and bones.

11. Progression rate 1%/year

WHAT YOU SEE

PROGRESSION RATE 1%/YEAR arrow

WHAT IT ENCODES

MGUS progresses to malignancy at $\sim 1\%$ /year (constant lifelong). Smoldering myeloma progresses much faster — about 10% /year in the first 5 years, then slowing — which is why SMM is watched more closely than MGUS. The steepening curve up the 'building' mirrors rising tumor burden and risk.

BOARD TRAP

WRONG answer: applying the MGUS 1% /year rate to smoldering myeloma surveillance intervals. The discriminator is that SMM's early progression ($\sim 10\%$ /year) demands tighter monitoring (e.g., every 2–3 months initially) than MGUS.

12. Solitary plasmacytoma

WHAT YOU SEE

SOLITARY PLASMACYTOMA, 1 LESION ONLY, RADIATION TARGET

WHAT IT ENCODES

Solitary plasmacytoma is a single clonal plasma cell mass in bone or soft tissue (extramedullary) with a NORMAL marrow ($<10\%$ clonal cells) and no CRAB. Treatment of choice is definitive local radiotherapy ($\sim 40\text{--}50$ Gy), which can be curative. Bone plasmacytomas progress to overt myeloma more often than extramedullary ones — follow with serial SPEP/FLC.

BOARD TRAP

WRONG answer: giving systemic myeloma chemotherapy for an isolated plasmacytoma. The discriminator is that it is localized with a clean marrow — first-line is radiation to the lesion, reserving systemic therapy for progression to multiple myeloma.

13. Nonsecretory myeloma

WHAT YOU SEE

NONSECRETORY MYELOMA, NO M SPIKE, FLC assay

WHAT IT ENCODES

Nonsecretory myeloma ($\sim 1\text{--}2\%$ of cases) has marrow plasmacytosis and CRAB end-organ damage but little or no detectable serum/urine M-protein on SPEP/UPEP. Most still have an abnormal serum free light chain ratio, so the FLC assay (plus marrow biopsy/imaging) makes the diagnosis. Mnemonic: 'silent spike, loud light chains.'

BOARD TRAP

WRONG answer: ruling out myeloma because SPEP/UPEP show no M-spike. The discriminator is nonsecretory/light-chain-only disease — order serum free light chains and a marrow biopsy; absence of an electrophoretic spike does NOT exclude myeloma.

14. All MM preceded by MGUS

WHAT YOU SEE

ALL MM PRECEDED BY MGUS banner

WHAT IT ENCODES

Essentially all multiple myeloma is preceded by an asymptomatic MGUS phase (shown by stored-sample studies), often progressing through smoldering myeloma — it is a biologic continuum, not a sudden event. This underlies the rationale for surveillance and for emerging early-intervention trials in high-risk precursors. Mnemonic: MGUS $\rightarrow$ SMM $\rightarrow$ MM is one escalator, not three separate buildings.

BOARD TRAP

WRONG answer: assuming myeloma arises de novo and that a prior 'benign' MGUS is unrelated. The discriminator is the established precursor continuum — a known MGUS patient who develops CRAB has progressed along that same clone, which is why baseline characterization and monitoring matter.